ClipSAT : EST I Math

Unit 0.01: EST I Literacy & Quantitative Diagnostic Drill

Name: _____

Q1. Consider the functions fx = 2x + 3 and gx = x + 7 . At what x -value do the graphs of fx and gx intersect?

[ ] A: 4
[ ] B: 7
[ ] C: 3
[ ] D: 10

Q2. The graph of the quadratic function fx = x^2 - 4x + 3 is shown below. What is the value of f0 ?

[ ] A: -3
[ ] B: -1
[ ] C: 1
[ ] D: 3

Q3. In the diagram below, lines AB and CD are parallel. Line segment EF is a transversal intersecting AB at G and CD at H. If the measure of angle angle AGH is 125^circ, what is the measure of angle angle GHD? text{Diagram description:} text{ Two parallel horizontal lines AB and CD are intersected by a transversal line EF. Points A and C are on the left, B and D are on the right. G is on AB, H is on CD. Angle AGH is in the upper left quadrant formed by the intersection.}

[ ] A: 55^circ
[ ] B: 125^circ
[ ] C: 90^circ
[ ] D: 180^circ

Q4. The bar chart below shows the number of students who prefer each of four different colors. What is the total number of students who prefer either blue or green?

[ ] A: 30
[ ] B: 40
[ ] C: 50
[ ] D: 60

Q5. Consider the functions fx = 2x + 3 and gx = x + 7 . At what x -value do the graphs of fx and gx intersect?

[ ] A: 4
[ ] B: 7
[ ] C: 3
[ ] D: 10

Q6. A recipe calls for 2 cups of flour for every 3 cups of sugar. If 8 cups of flour are used, how many cups of sugar are needed?

[ ] A: 10 cups
[ ] B: 11 cups
[ ] C: 12 cups
[ ] D: 13 cups

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Q7. If 9^{x+1} = 27^{x-1} , what is the value of x ?

[ ] A: 2
[ ] B: 3
[ ] C: 5
[ ] D: 7

Q8. The graph of the quadratic function fx = x^2 - 4x + 3 is shown below. What is the value of f0 ?

[ ] A: -3
[ ] B: -1
[ ] C: 1
[ ] D: 3